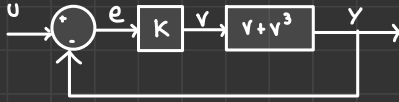


SISTEMA NON LINEARE STATICO

EQUAZIONE DIFFERENZIALE NON DIPENDENTE DAL TEMPO



$$e = u - y \quad v = Ke \quad v + v^3 = y$$

SOSTITUIAMO

$$Ke + (Ke)^3 = y \rightarrow K(u - y) + K^3(u - y)^3 = y$$

SUPPONENDO $K > 0$ DIVIDIAMO PER K^3

$$\frac{K(u - y) - y}{K^3} + (u - y)^3 = 0$$

FACCIAMO $K \rightarrow \infty$

$$(u - y)^3 = 0 \rightarrow u = y$$

EQUAZIONI DIFFERENZIALI LINEARI

$$y^k + a_{k-1}(t)y^{k-1} + \dots + a_0(t)y = g(t)$$

Y FUNZIONE

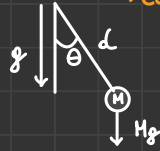
$$g(t) = 0 \quad \text{OMOGENA}$$

$$a(t) = a \quad \text{A COEFFICIENTI COSTANTI}$$

$$y(t) = y \quad \text{AUTONOMA}$$

SISTEMA NON LINEARE DINAMICO

EQUAZIONE DIFFERENZIALE DIPENDENTE DAL TEMPO



$$U(t) = M d^2 \ddot{\theta} + M g d \sin(\theta)$$

AGGIUNGIAMO UN FEEDBACK CON UN GUADAGNO

MOLTO ELEVATO (E MOLTO PICCOLO)

$$U(t) = \frac{1}{\epsilon} (v - \dot{\theta})$$

$$M d^2 \ddot{\theta} + M g d \sin(\theta) = \frac{1}{\epsilon} (v - \dot{\theta})$$

MOLTIPLICO PER ϵ

$$\epsilon (M d^2 \ddot{\theta} + M g d \sin(\theta)) = (v - \dot{\theta}) \quad \epsilon \rightarrow 0$$

$$v - \dot{\theta} = 0 \rightarrow v = \dot{\theta}$$

EDUAZIONE DIFFERENZIALE DI PRIMO ORDINE

$$\dot{y} = a(t)y + b(t)$$

se $b=0$ OMogenea

SOLUZIONE:

$$y(t) = e^{A(t)} \int e^{-A(t)} b(t) dt + c_0$$

↳ TERMINE FORZANTE

↳ PENSANDO A UN SISTEMA DI
CONTROLLI È L'INGRESSO

$$A(t) = \int_{t_0}^t a(\tau) d\tau$$

↳ QUALSIASI PRIMITIVA $a(t)$

DEFINIAMO IL PROBLEMA DI CAUCHY:

$$\begin{cases} \dot{y}(t) = a(t)y + b(t) \\ y(t_0) = y_0 \end{cases}$$

TROVANDO LA SOLUZIONE

$$y(t) = \underbrace{e^{A(t)} y_0}_{\text{EVOLUZIONE LIBERA}} + \underbrace{\int_0^t e^{A(t)-A(\tau)} b(\tau) d\tau}_{\text{EVOLUZIONE FORZATA}}$$

EVOLUZIONE
LIBERA

↓
DIPENDE DALL'
USCITA E NON
DALL'INGRESSO

↳ EVOLUZIONE FORZATA

↳ DIPENDE DAL FORZAMENTO

NOTIAMO CHE:

$$a(t) = a \rightarrow A(t) = a(t - t_0)$$

DI CONSEGUENZA

$$y(t) = e^{a(t-t_0)} y_0 + \int_0^t e^{a(t-\tau)} b(\tau) d\tau = e^{a(t-t_0)} y_0 + e^t * b(t)$$

CAMBIAAMENTO DI VARIABILE INGRESSO - STATO - USCITA

RAPPRESENTAZIONE INGRESSO - STATO - USCITA

$$\begin{cases} \dot{x} = Ax + Bu \\ y = Cx \end{cases}$$

$$x = Tz$$

$$\begin{cases} \dot{z} = \tilde{A}z + \tilde{B}u \\ y = \tilde{C}z \end{cases}$$

$$\tilde{A} = T^{-1}AT \quad \tilde{B} = T^{-1}B \\ \tilde{C} = CT$$

TRASFORMATA DI LAPLACE DEL COSENO

$$\begin{aligned} \mathcal{L}\{\cos(\omega t)\} &= \int_0^\infty \cos(\omega t) e^{-st} dt = \int_0^\infty \left(\frac{e^{j\omega t} + e^{-j\omega t}}{2} \right) e^{-st} dt \\ &= \int_0^\infty e^{(j\omega - s)t} dt + \int_0^\infty e^{(-j\omega - s)t} dt \\ &= \frac{1}{2} \left(\frac{1}{s - j\omega} + \frac{1}{s + j\omega} \right) = \frac{(s + j\omega) + (s - j\omega)}{2(s - j\omega)(s + j\omega)} = \frac{s}{(s^2 + \omega^2)} \end{aligned}$$

ANTITRASFORMATA DI LAPLACE

$$F(s) = \frac{N(s)}{D(s)} = \frac{\sum b_i s^i}{\sum a_i s^i} \rightarrow \begin{matrix} \text{ZERI} \\ \text{POLI} \end{matrix}$$

POLI SINGOLI

$$F(s) = \frac{N(s)}{(s-p_1) \dots (s-p_m)} = h_0 + \frac{h_1}{(s-p_1)} + \dots + \frac{h_m}{(s-p_m)} = h_0 + \sum_{i=1}^m \frac{h_i}{(s-p_i)}$$

$$h_i = \lim_{s \rightarrow p_i} (s-p_i) \frac{N(s)}{D(s)}$$

$$f(t) = \sum_{i=1}^m h_i e^{p_i t}$$

POLI MULTIPLI

$$F(s) = \frac{N(s)}{(s-p_1)^{r_1} \dots (s-p_m)^{r_m}} = \sum_{i=1}^m \sum_{j=1}^{r_i} \frac{h_{ij}}{(s-p_i)^{j-1}}$$

$$h_{ij} = \lim_{s \rightarrow p_i} \frac{1}{(j-1)!} \frac{d^{j-1}}{ds^{j-1}} (s-p_i)^{j-1} \frac{N(s)}{D(s)}$$

$$f(t) = \sum_{i=1}^m \sum_{j=1}^{r_i} \frac{h_{ij}}{(j-1)!} t^{j-1} e^{p_i t}$$

MODI PROPRI DEL SISTEMA

1. POLI D.C.

h COSTANTE

$h e^{pt}$ ESPONENZIALE

$h e^{\sigma t} \sin(\omega t + \varphi)$ OSCILLATORIO

$$Y(s) = \frac{N(s)}{(s-p_1) \dots (s-p_n)}$$

$$Y(s) = \frac{1}{(s-p)} \quad Y(t) = h e^{pt} \quad \uparrow \text{ANDAMENTO NEL TEMPO IN FUNZIONE DI } p$$

$p < 0$ CONVERGENTE

$p = 0$ COSTANTE

$p > 0$ DIVERGENTE

MODI OSCILLATORIO

$$Y(s) = \frac{h}{(s-p)} + \frac{h^*}{(s-p)^*} = \frac{h}{s-\sigma-j\omega} + \frac{h^*}{s-\sigma+j\omega}$$

$$Y(t) = h e^{(\sigma+j\omega)t} + h^* e^{(\sigma-j\omega)t}$$

$$h = |h| e^{j\varphi}$$

$$Y(t) = 2|h| e^{\sigma t} \sin(\omega t + \varphi)$$

$\sigma < 0$ CONVERGENTE

$\sigma = 0$ STAZIONARIO

$\sigma > 0$ DIVERGENTE

$$y^{(n)} + a_{n-1} y^{(n-1)} + \dots + a_0 = b_0 u(t) + \dots + b_m u^{(m)}(t)$$

$$L \left\{ \frac{d^m}{dt^m} y(t) \right\} = s^m Y(s) + C_1 y^{(m-1)}(s) + \dots + C_{m-1} s^{m-1} Y(s)$$

$$Y(s) = \frac{b_n s^n + \dots + b_0}{a_m s^m + \dots + a_0} U(s) + \frac{C_1 v(s)}{a_m s^m + \dots + a_0}$$

$$Y(s) = Y_2(s) - Y_1(s) = \underline{F(s)} U(s) - Y_1(s)$$

$F(s)$ FUNZIONE DI TRASFERIMENTO

DA PPOCINO TUA!

C1 PERMETTE DI OSSERVARE:

- STABILITA' ASINTOTICA

- VELOCITA' DI CONVERGENZA

- **COMPORTAMENTO OSCILLATORIO**

$$F(s) = \frac{N(s)}{D(s)} \quad u(t) = \frac{t^k}{k!}$$

$$Y(s) = \frac{N(s)}{D(s)} U(s)$$

$$Y(z) = \underbrace{L^{-1} \left[\sum_{i=1}^{n_{a,s}} \frac{p_i}{(s-p_i)} \right]}_{A.S.} + \underbrace{L^{-1} \left[\sum_{j=1}^{n_{s,s}} \frac{\omega_j}{(s-p_j)} \right]}_{S.S.} + \underbrace{L^{-1} \left[\sum_{k=1}^m \frac{h_k}{(s-p_k)} \right]}_{INST.}$$

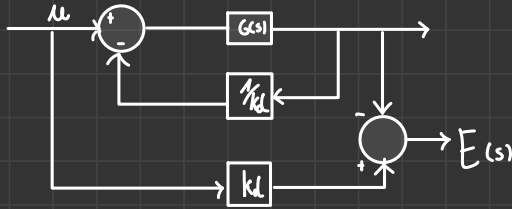
$$\lim_{t \rightarrow \infty} y(t) = \lim_{s \rightarrow 0} s F(s)$$

$$a_m y^{(m)} + \dots + a_0 = (s-p)(s-p)^q = s^2 + 2\sigma s + \omega^2 + \sigma^2 = s^2 + 2 \int \omega_m s + \omega_m^2$$

ω_m PULSAZIONE NATURALE = $\sqrt{\omega_0}$

$$\text{Smorzamento} = \frac{Q_1}{2W_m}$$

COMPORTAMENTO A REGIME: CLASSIFICAZIONE DEI TIPI



$$y = y_d \quad \lim_{t \rightarrow \infty} e(t) = 0$$

CI SPOSTIAMO SUL PIANO S

$$E(s) = -W(s)U(s) + K_d U(s)$$

$$= \left(K_d - \frac{G(s)}{1 + \frac{G(s)}{K_d}} \right) U(s) = \left(K_d - \frac{K_d G(s)}{K_d + G(s)} \right) U(s) = \frac{K_d^2}{K_d + G(s)} U(s)$$

↳ FUNZIONE DI TRASFERIMENTO INGRESSO-CARORE $\frac{e^k}{k!} = \frac{1}{s^{k+1}}$

APPLICHIAMO IL TEOREMA DEL VALORE FINITE

$$\lim_{t \rightarrow \infty} e(t) = \lim_{s \rightarrow 0} s E(s) = \lim_{s \rightarrow 0} s \frac{K_d^2}{K_d + G(s)} \frac{1}{s^{k+1}} = \lim_{s \rightarrow 0} \frac{K_d^2}{K_d + \frac{G(s)}{s^h}} \frac{1}{s^k} = \lim_{s \rightarrow 0} \frac{K_d^2}{s^h K_d + G(s)} \cdot \frac{1}{s^{h-k}}$$

$$G(s) = P(s) \cdot C(s)$$

$$G(s) = \frac{G'(s)}{s^h}$$

$$G'(0) = K_6$$

$$h > k \quad e(\infty) = 0$$

$$h < k \quad e(\infty) = \infty$$

$$h = k \quad \begin{cases} \rightarrow h=0 \\ \rightarrow h \geq 1 \end{cases}$$

$$\frac{K_d^2}{K_d + K_6}$$

$$\frac{K_d^2}{K_6}$$

$$h \backslash k$$

$$0$$

$$1$$

$$2$$

$$\overline{\sigma}_1(t)$$

$$\frac{K_d^2}{K_d + K_6}$$

$$0$$

$$0$$

$$\overline{\sigma}_2(t)$$

$$\infty$$

$$\frac{K_d^2}{K_6}$$

$$0$$

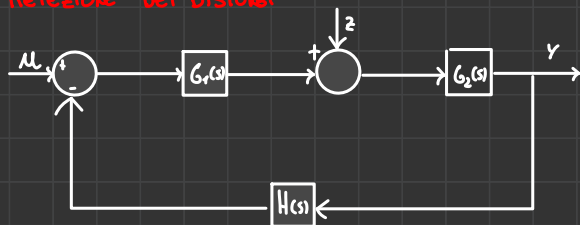
$$\overline{\sigma}_3(t)$$

$$\infty$$

$$\infty$$

$$\frac{K_d^2}{K_6}$$

REIEZIONE DEI DISTURBI



$$W_z(s) = \frac{G_2(s)}{1 + G_1(s)G_2(s)H(s)} \quad Z(s) = \frac{1}{s^{k+1}}$$

$$\lim_{s \rightarrow 0} s W_z(s) Z(s) = \lim_{s \rightarrow 0} \frac{G_2(s)}{1 + G_1(s)G_2(s)H(s)} \cdot \frac{1}{s^{k+1}}$$

$$G_1(s) = \frac{G'_1(s)}{s^h}$$

$$\lim_{s \rightarrow 0} \frac{G_2(s)}{1 + G_1(s)G_2(s)H(s)} \cdot \frac{1}{s^k} = \lim_{s \rightarrow 0} \frac{G_2(s)}{s^h + G_1(s)G_2(s)H(s)} \cdot \frac{1}{s^{h-k}} \quad G'_1(0) = k_{G_1}$$

$$G_2(0) = k_{G_2}$$

$$h > k \quad e_z(\infty) = 0$$

$$h < k \quad e_z(\infty) = \infty$$

$$h = k \quad e_z(\infty) = \frac{k_{G_2}}{k_{G_1} k_{G_2}} \cdot \frac{1}{k_{G_1}}$$

RISPOSTA ARMONICA

$$u \rightarrow \boxed{G(s)} \rightarrow y \quad u(t) = \sin(\omega t)$$

$$Y(s) = G(s) \cdot \frac{\omega}{s^2 + \omega^2}$$

DECOMPOSIZIONE IN POLI E RESIDUI

$$Y(s) = \frac{R_1}{s - p_1} + \frac{R_2}{s - p_2} + \frac{R_3}{s - p_3} + \frac{R_3^*}{(s - p_3)^*}$$

STUDIO LA RISPOSTA A REGIME PERMANENTE

$$R = \lim_{s \rightarrow j\omega} (s - j\omega) G(s) \frac{R}{(s - j\omega)(s + j\omega)} = \frac{G(j\omega)}{2j}$$

$$R^* = \frac{G(j\omega)^*}{-2j}$$

$$G(j\omega) = |G(j\omega)| e^{j\varphi}$$

$$\mathcal{L}^{-1} \left\{ \frac{R}{(s - j\omega)} \right\} = R e^{j\omega t}$$

$$\mathcal{L}^{-1} \left\{ \frac{R^*}{(s + j\omega)} \right\} = R^* e^{-j\omega t}$$

$$y_p(t) = \frac{|G(j\omega)| e^{j\varphi}}{2j} \cdot e^{j\omega t} - \frac{|G(j\omega)| e^{j\varphi}}{2j} e^{-j\omega t} = |G(j\omega)| \sin(\omega t + \varphi)$$

RISPOSTA ARMONICA

$$\xrightarrow{\text{MIS}} \boxed{G(s)} \xrightarrow{Y(s)}$$

$$u(t) = \sin(\omega t)$$

$$U(s) = \frac{\omega}{s^2 + \omega^2}$$

$$Y(s) = \frac{h_1}{s-p_1} + \frac{h_2}{s-p_2} + \frac{h_3}{s-p_3} + \frac{h_3^*}{(s-p_3)^*}$$

$$h = \lim_{s \rightarrow j\omega} (s - j\omega) G(s) \frac{\omega}{(s - j\omega)(s + j\omega)} = \frac{G(j\omega)}{2j}$$

$$h^* = \frac{G(-j\omega)}{-2j}$$

$$G(j\omega) = |G(j\omega)| e^{j\phi}$$

$$\mathcal{L}^{-1} \left\{ \frac{h}{s - j\omega} \right\} = h e^{j\omega t} \quad \mathcal{L}^{-1} \left\{ \frac{h^*}{s + j\omega} \right\} = h^* e^{-j\omega t}$$

$$y_p(t) = h e^{j\omega t} + h^* e^{-j\omega t} = \frac{|G(j\omega)| e^{j\phi}}{2j} e^{j\omega t} + \frac{|G(j\omega)| e^{-j\phi}}{-2j} e^{-j\omega t} = |G(j\omega)| \sin(\omega t + \phi)$$

DIAGRAMMA DI BODE

$$G(s) = K_G s^h \frac{\left(\frac{s^2}{\omega_n^2} + \frac{2\zeta}{\omega_n} s + 1 \right) (s + 1)}{\left(\frac{s^2}{\omega_m^2} + \frac{2\zeta}{\omega_m} s + 1 \right) (s + 1)}$$

GUARDARE COSTANTE

$$|K_G|_{dB} = 20 \log_{10} |K_G|$$

$$\angle K_G \rightarrow K_G > 0 \rightarrow 0^\circ$$

$$K_G < 0 \rightarrow -180^\circ$$

ZERI E POLI ALL'ORIGINE

$$s^h \xrightarrow{s=j\omega} (j\omega)^h$$

$$|(j\omega)^h|_{dB} = 20h \log_{10}(\omega) = 20h \lambda$$

$$\angle (j\omega)^h = 90h^\circ$$

$$\lambda = \log_{10} \omega \rightarrow \omega = 10^\lambda$$

ZERI E POLI REALI

$$(s\tau + 1) \quad S = s\omega$$

$$|(s\omega\tau + 1)|_{dB} = \sqrt{(s\omega\tau)^2 + 1} \quad dB = 10 \log_{10}((\omega\tau)^2 + 1)$$

$\omega\tau \gg 1$

$$10 \log_{10}(\omega\tau)^2 = 20 \log_{10}(\omega\tau) = 20 \log_{10}(\omega) + 20 \log_{10}(\tau) \approx 20 \log_{10}(\omega) = 20 \lambda$$

$\omega\tau \ll 1$

$$10 \log_{10}(1) = 0$$

$\omega\tau \approx 1$

$$\begin{aligned} \angle(s\omega\tau + 1) \quad \omega\tau \gg 1 &\approx \angle(1) = 90^\circ \\ \omega\tau \ll 1 &\approx \angle(s\omega\tau) = 0^\circ \\ \omega\tau \approx 1 &\approx \angle(s+1) = 45^\circ \end{aligned}$$

ZERI E POLI COMPLEGGATI

$$S = s\omega$$

$$\left(\frac{2s\omega}{\omega_m} s + 1 - \frac{\omega^2}{\omega_m^2} \right)$$

Calcolo IL MODULO

$$\begin{aligned} &\left| \left(\frac{2s\omega}{\omega_m} s + 1 - \frac{\omega^2}{\omega_m^2} \right) \right|_{dB} \\ &= 10 \log_{10} \left(\left(\frac{2s\omega}{\omega_m} s \right)^2 + \left(1 - \frac{\omega^2}{\omega_m^2} \right)^2 \right) \end{aligned}$$

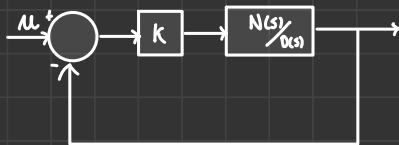
$\omega \gg \omega_m$

$$\begin{aligned} &= 10 \log_{10} \left(\frac{\omega^2}{\omega_m^2} \right)^2 = 40 \log_{10} \left(\frac{\omega}{\omega_m} \right) = 40 \log_{10}(\omega) - 40 \log_{10}(\omega_m) \\ &\sim 40 \log_{10}(\omega) = 40 \lambda \end{aligned}$$

$$\omega \ll \omega_m \quad \approx 11 \approx 0 \text{ dB}$$

$$\omega = \omega_m \quad 20 \log_{10}(2) \approx 6 \text{ dB}$$

SISTEMI A FASE NON MINIMA

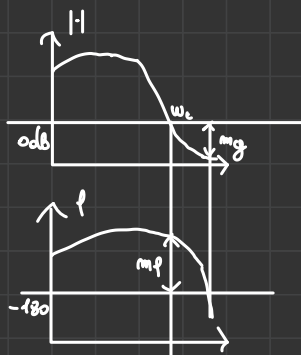
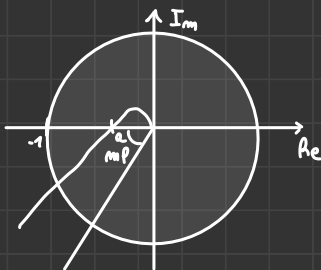


$$W(s) = \frac{k \cdot \frac{N(s)}{D(s)}}{1 + k \cdot \frac{N(s)}{D(s)}} = \frac{k \cdot N(s)}{D(s) + k \cdot N(s)}$$

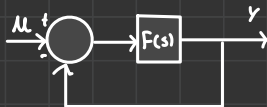
$$K \rightarrow 0 \quad k \cdot N(s) \rightarrow 0$$

$$K \rightarrow \infty \quad k \cdot N(s) \rightarrow \infty$$

MARGINI DI STABILITÀ



CRITERIO DI NYQUIST



$$W(s) = \frac{F(s)}{1 + F(s)} \quad F(s) = \frac{N(s)}{D(s)}$$

$$\downarrow$$

$$= \frac{\frac{N(s)}{D(s)}}{1 + \frac{N(s)}{D(s)}} = \frac{N(s)}{D(s) + N(s)}$$

$$1 + F(s) = 1 + \frac{N(s)}{D(s)} = \frac{D(s) + N(s)}{D(s)}$$

$$R_{(1+F(s), 0)} = m_{\text{ZERI P.A.P.}}^0 (1 + F(s)) - m_{\text{POLI P.A.P.}}^0 (1 + F(s))$$

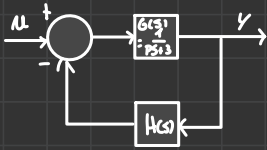
$$= m_{\text{POLI P.A.P.}} W(s) - m_{\text{POLI P.A.P.}} F(s)$$

CRITERIO DI STABILITÀ DI NYQUIST

$$R_{(1+F), 0} = -m_{\text{POLI P.A.P.}} F(s) \longrightarrow R_{(F), -1} = -m_{\text{POLI P.A.P.}}^0 F(s)$$

FUNZIONE DI SENSIBILITA'

DEF: $S_{\alpha}^Q = \frac{\frac{dQ}{d\alpha}}{\frac{Q}{\alpha}} = \frac{\alpha}{Q} \frac{dQ}{d\alpha}$

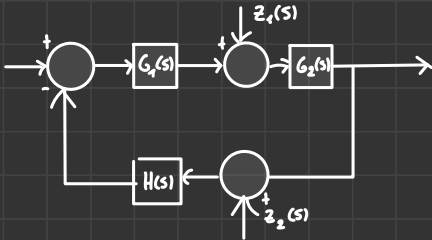


$$W(s) = \frac{G(s)}{1 + G(s)H(s)}$$

$$S_p^W = \frac{p}{W} \frac{dW}{dp} = \left(\frac{p}{W} \frac{dW}{dp} \right) \left(\frac{G}{G} \cdot \frac{dG}{dG} \right) = \left(\frac{p}{G} \cdot \frac{dG}{dp} \right) \left(\frac{G}{W} \frac{dW}{dG} \right) = S_p^G S_G^W$$

$$S_G^W = \frac{G(s)}{W(s)} \cdot \frac{dW(s)}{dG(s)} = \frac{1}{1 + G(s)H(s)}$$

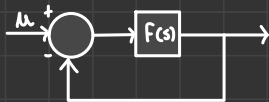
$$S_H^W = \frac{H(s)}{W(s)} \cdot \frac{dW(s)}{dH(s)} = \frac{G(s)H(s)}{1 + G(s)H(s)}$$



$$Y_{Z_1}(s) = \frac{G_2(s)}{1 + G_1(s)G_2(s)H(s)} Z_1(s) = S_G^W \cdot G_2(s) \cdot Z_1(s)$$

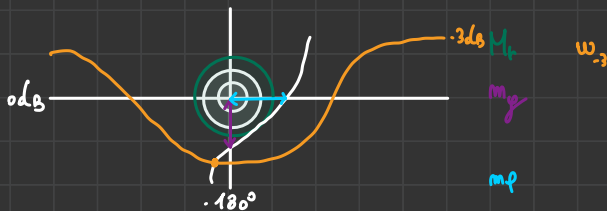
$$Y_{Z_2}(s) = \frac{G_1(s)G_2(s)H(s)}{1 + G_1(s)G_2(s)H(s)} Z_2(s) = S_H^W \cdot Z_2(s)$$

CARTO DI NICHOLS

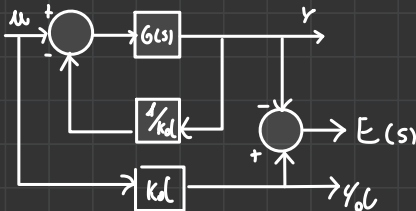


$$W(s) = \frac{F(s)}{1 + F(s)}$$

$$F(s) = \frac{W(s)}{1 - W(s)}$$



ERRORE DI RIPRODUZIONE DI UNA SINUSOIDE E SPECIFICHE IN FREQUENZA



$$E(s) = Y_d(s) - Y(s) = K_d U(s) - \frac{K_d G(s)}{K_d + G(s)} U(s) = \left(\frac{K_d^2}{K_d + G(s)} \right) U(s)$$

$$Y_d(s) = K_d U(s)$$

$$Y(s) = W(s) U(s)$$

$$W(s) = \frac{G(s)}{1 + \frac{G(s)}{K_d}} = \frac{K_d G(s)}{K_d + G(s)}$$

$$\frac{E(s)}{U(s)} = \frac{K_d}{1 + \frac{G(s)}{K_d}} = \frac{K_d}{1 + F(s)}$$

$$\frac{G(s)}{K_d} = F(s)$$

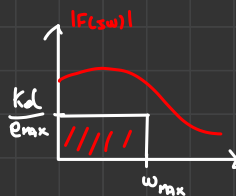
$$\left| \frac{E(s)}{U(s)} \right| = \frac{K_d}{|1 + F(s)|} \sim \frac{K_d}{|F(s)|}$$

$$e = \frac{K_d}{|F(s)|}$$

$$e \leq e_{\max} \quad \omega \leq \omega_{\max}$$

$$e_{\max} \geq \frac{K_d}{|F(s)|}$$

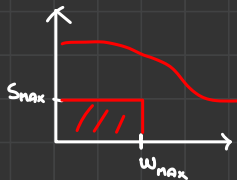
$$|F(s)| \geq \frac{K_d}{e_{\max}}$$



RISERVAZIONE DI DISTURBI SINUSOIDALI

$$S_G^W = \frac{1}{1 + H(s)G(s)} \quad H(s) \cdot G(s) = F(s)$$

$$= \frac{1}{|1 + F(s)|} \leq S_{\max} \sim \frac{1}{|F(s)|} \leq S_{\max} = |F(s)| \geq \frac{1}{S_{\max}}$$



SISTEMI NON LINEARI E LINEARIZZAZIONE INTORNO A UN PUNTO D'EQUILIBRIO

$$y = \sin(x) \quad y = \sin(x_0) + \frac{d \sin(x_0)}{dx_0} (x - x_0) + \dots$$



$$M d^2 \ddot{\theta} = -M g d \sin(\theta) - D \dot{\theta} + \tau \rightarrow \underbrace{M d^2 \ddot{\theta}}_{\text{INERZIA}} + \underbrace{M g d \sin(\theta)}_{\text{FORZA}} + \underbrace{D \dot{\theta}}_{\text{ATTRITO}} = \underbrace{\tau}_{\text{COPPIA IMPOSTA}}$$

PER RISOLVERE IL PROBLEMA IDENTIFICO UN ←

PUNTO D'EQUILIBRIO IN θ_0

$$\theta = \theta_0 + \Delta\theta$$

ESPLICITO GLI ALTRI θ DEL SISTEMA

$$\dot{\theta} = \dot{\Delta\theta} \quad \ddot{\theta} = \ddot{\Delta\theta} \quad \sin(\theta) = \sin(\theta_0) + \cos(\theta_0) \Delta\theta$$

←
L'EQUAZIONE
DIVENTA

$$M d^2 \ddot{\Delta\theta} + D \dot{\Delta\theta} + M g d (\sin(\theta_0) + \cos(\theta_0) \Delta\theta) = \tau$$

↓ NEL PUNTO D'EQUILIBRIO IL SISTEMA È FERMO
LE DERIVATE VALGONO 0

$$M g d (\sin(\theta_0)) = \tau_0$$

τ_0 = COPPIA PER TENERE FERMO IL SISTEMA

$$\tau = \tau_0 + \Delta\tau$$

IL SISTEMA INIZIALE DIVENTA

$$M_d l^2 \Delta \ddot{\theta} + D \Delta \dot{\theta} + M_g d (\sin(\theta_0) + \cos(\theta_0) \Delta \theta) = \tau_0 + \Delta\tau$$

EQUAZIONE LINEARIZZATA

INTORNO ALLA POSIZIONE D'EQUILIBRIO (θ_0, τ_0)

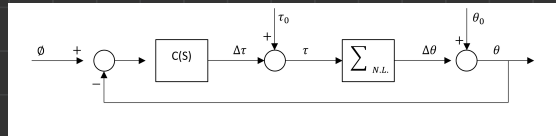
↓
APPLICO LA TRASFORMATA
DI LAPLACE

FUNZIONE DI TRASFERIMENTO DEL NOSTRO SISTEMA LINEARIZZATO:

$$\frac{\Delta\theta}{\Delta\tau} = \frac{1}{M_d l^2 s^2 + D s + M_g d (\cos(\theta_0))} = P(s)$$

PRENDE $\Delta\tau$ PRODUCE $\Delta\theta$
LINEARE

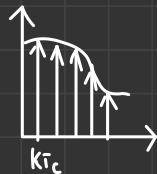
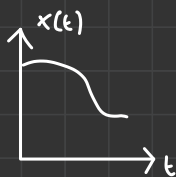
PRENDE τ PRODUCE θ
NON LINEARE



LE TRAIETTORIE DEL SISTEMA NON LINEARE CONVERGONO
ASINTOTICAMENTE AL PUNTO D'EQUILIBRIO CONSIDERATO
SE IL SISTEMA LINEARIZZATO ASSOCIATO È ASINTOTICAMENTE
STABILE

TEOREMA DEL CAMPIONAMENTO

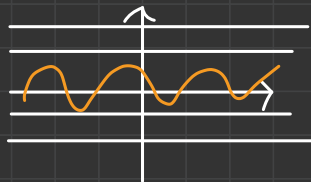
$$f_c \min > 2 f_{\max}$$



$$x_c(t) = \sum_{k=-\infty}^{\infty} x(t) \delta(t - kT_c)$$

$$X_c(s) = \sum_{k=-\infty}^{\infty} x(t) e^{-skT_c}$$

$$X_c(s) = \sum_{k=-\infty}^{\infty} x(t) e^{-\sigma k \frac{2\pi}{\omega_c}} e^{-j\omega k \frac{2\pi}{\omega_c}} \quad T_c = \frac{2\pi}{\omega_c}$$



$$f_c = 1/T_c$$

$$\omega_c = 2\pi f_c$$

$$S = \sigma + j\omega$$

$$f(t) = \frac{a_0}{2} \sum_{k=1}^{\infty} [a_k \cos(k\omega_c t) + b_k \sin(k\omega_c t)]$$

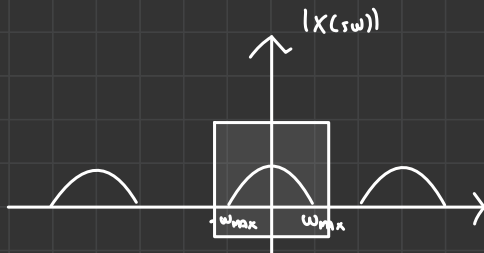
$$f(t) = \sum_{k=-\infty}^{\infty} c_k e^{-k j \omega_c t}$$

$$c_k = \frac{1}{T_c} \int_{-T_c/2}^{T_c/2} \delta(t - kT_c) e^{-k j \omega_c t} dt = \frac{1}{T_c}$$

$$X_c(t) = \sum_{k=-\infty}^{\infty} \frac{1}{T_c} e^{-k j \omega_c t}$$

$$x_c(t) = x(t) \sum_{k=-\infty}^{\infty} \frac{1}{T_c} e^{-k j \omega_c t} = \frac{1}{T_c} \sum_{k=-\infty}^{\infty} x(t) e^{-k j \omega_c t}$$

$$X_c(s) = \frac{1}{T_c} \sum_{k=-\infty}^{\infty} X(s - j k \omega_c) \quad S = \sigma + j\omega$$



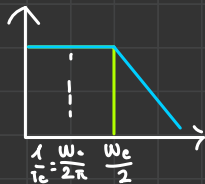
ORGANO DI TENUTA



$$X_{\text{GRADINO}}(t) = \sum_{k=0}^{\infty} x(k) \delta(t - kT_c) - x(k) \delta(t - (k+1)T_c)$$

$$X_{\text{GRADINO}}(s) = \sum_{k=0}^{\infty} x_k \frac{1}{s} e^{-skT_c} - x_k \frac{1}{s} e^{-s(k+1)T_c} = \left(\sum_{k=0}^{\infty} x(k) e^{-skT_c} \right) \left(\frac{1 - e^{-sT_c}}{s} \right)$$

$X(s)$ $T(s)$



$$e^x = 1 + x$$

$$e^{sT_c} = 1 + sT_c$$

$$T(s) = \frac{1 - \frac{1}{1+sT_c}}{s} = \frac{T_c}{1+sT_c}$$

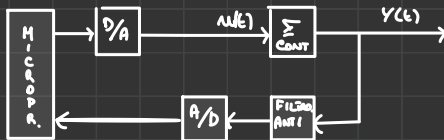
SCELTA DEL TEMPO DI CAMPIONAMENTO

$$\omega_c > 2\omega_{\max}$$

$$\omega_c = 2\pi \frac{1}{T_c}$$

$$2\pi \frac{1}{T_c} > 2\omega_{\max}$$

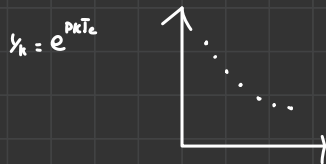
$$\frac{1}{T_c} > \frac{\omega_{\max}}{\pi} \rightarrow T_c > \frac{\pi}{\omega_{\max}}$$



TRANSFORMATA Z

ES. $y_{k+1} = 3y_k + 10y_k$

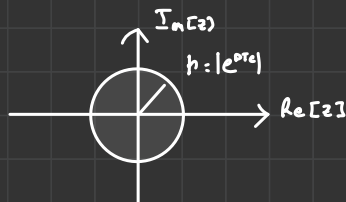
$$Z\{y_k\} = Y_k(z) = \sum_{k=0}^{\infty} y_k z^{-k} = \frac{1}{1 - y_k z^{-k}} \quad |y_k z^{-k}| < 1$$



$$Y_k(z) = \sum_{k=0}^{\infty} e^{pkT_c} z^{-k} = \sum_{k=0}^{\infty} (e^{pT_c} \cdot z^{-1})^k = \frac{1}{1 - e^{pT_c} z^{-1}} = \frac{z}{z - e^{pT_c}}$$

$$|e^{pT_c} z^{-1}| < 1$$

$$|z| > e^{pT_c}$$



$$x_c(t) = \sum_{k=0}^{\infty} x(kT_c) \delta(t - kT_c)$$

$$X_c(s) = \sum_{k=0}^{\infty} x(kT_c) e^{-s k T_c} \quad z = sT_c$$

$$X_c(z) = \sum_{k=0}^{\infty} x_k z^{-k}$$

TRASFORMATA ESATTA DA $F(s)$ A $F(z)$



$$C(s) = \frac{1}{s-p} \quad \mathcal{L}^{-1} \left\{ \frac{1}{s-p} \right\} = e^{pt} = e(t)$$

$$e_c(t) = \sum_{k=0}^{\infty} e^{p k T_c} \delta(t - k T_c)$$

$$C(z) = \sum_{k=0}^{\infty} e^{p k T_c} z^{-k} = \frac{1}{1 - e^{p T_c}} z^{-1} = \frac{z}{z - e^{p T_c}}$$

$$C(s) = \frac{h_1}{s-p_1} + \frac{h_2}{s-p_2} \dots$$

$$C(z) = \frac{h_1}{z - e^{p_1 T_c}} + \dots$$

$$K_{c_s} = \lim_{s \rightarrow 0} s \cdot \frac{1}{s-p} \cdot \frac{1}{s} = -\frac{1}{p}$$

$$K_{c_z} = \lim_{z \rightarrow 1} (1 - z^{-1}) \frac{1}{z - e^{p T_c}} \frac{1}{1 - z^{-1}} = \frac{1}{1 - e^{p T_c}}$$

MAPPING DAL PIANO S AL PIANO Z

$$X_c(s) = \sum_{k=0}^{\infty} x(kT_c) e^{-skT_c}$$

$$z = e^{sT_c}$$

$$X_k(z) = \sum_{k=0}^{\infty} x(kT_c) z^{-k}$$

$$s = \sigma + j\omega$$

$$z = e^{sT_c} = e^{\sigma T_c} e^{j\omega T_c}$$

$$T_c = 2\pi/\omega_c$$

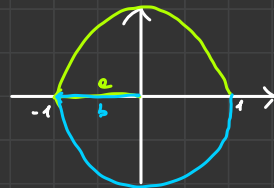
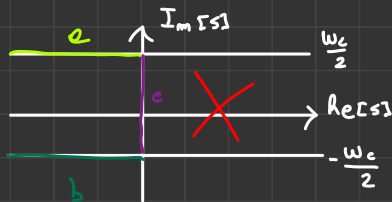
$$= e^{\sigma \frac{2\pi}{\omega_c}} e^{j\omega \frac{2\pi}{\omega_c}}$$

$$z = e^{j\omega \frac{2\pi}{\omega_c}}$$

$$z = e^{j\left(\frac{\omega_c}{2}\right) \frac{2\pi}{\omega_c}} = e^{j\pi}$$

$$z = e^{j\left(\frac{\omega_c}{2}\right) \frac{2\pi}{\omega_c}} = e^{j\pi}$$

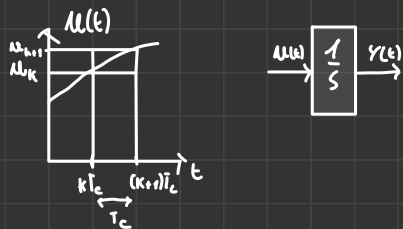
$$z = e^{j\pi} = -1$$



$$|z| = \sqrt{\sigma^2 + \omega^2}$$

$$|z| \geq 1$$

TRASFORMATE APPROSSIMATE DAL PIANO S AL PIANO Z



1. DIRETTO

$$y_{k+1} = y_k + u_k T_c$$

$$Y(z)(1 - z^{-1}) = T_c U(z) z^{-1}$$

$$\frac{Y(z)}{U(z)} = \frac{T_c z^{-1}}{1 - z^{-1}} \approx \frac{1}{s} = \frac{T_c z^{-1}}{1 - z^{-1}} \rightarrow S = \frac{1 - z^{-1}}{T_c z^{-1}}$$

2. ECCESSO

$$y_{k+1} = y_k + u_{k+1} T_c$$

$$Y(z)(1 - z^{-1}) = T_c U(z)$$

$$\frac{Y(z)}{U(z)} = \frac{T_c}{1 - z^{-1}} \approx \frac{1}{s} \rightarrow \frac{1 - z^{-1}}{T_c} \rightarrow S = \frac{T_c}{1 - z^{-1}}$$

3. FORMA TRAPEZOIDALE

$$y_{k+1} = y_k + \left(u_k + u_{k+1} \right) \frac{T_c}{2}$$

$$Y(z)(1 - z^{-1}) = \frac{T_c}{2} U(z)(1 + z^{-1})$$

$$\frac{Y(z)}{U(z)} = \frac{T_c}{2} \frac{(1 + z^{-1})}{1 - z^{-1}} \approx \frac{1}{s} \rightarrow S = \frac{2}{T_c} \frac{1 - z^{-1}}{1 + z^{-1}}$$

